Soft-Guided Open-Vocabulary Semantic Segmentation of Remote Sensing Images

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***Abstract*—Open-vocabulary remote sensing (RS) semantic segmentation strives to assign both seen and unseen class labels to individual pixels in RS images. Existing models follow the "fine-tune" paradigm based on vision-language models (VLMs). However, as VLMs are predominantly tailored to natural scenes, these directly fine-tuned models often collapse into the seen categories and show insensitivity in perceiving RS semantic cues. This critical issue of model collapse is closely related to the misalignment between image and text, making them struggle with the unique challenges of RS images, such as complex and diverse scenes, and objects with significant scale differences. To this end, we propose a soft-guided open-vocabulary RS semantic segmentation framework, which is the first to explore how to softly adapt VLMs to the downstream task of semantic segmentation for RS images. Concretely, instead of directly fine-tuning, we introduce a generalization compensation strategy, which employs an additional frozen VLM encoder to provide implicit semantic guidance for dynamic optimization of visual representation. By introducing prior knowledge from the frozen encoder, this soft strategy compensates potential losses incurred during fine-tuning, thus enhancing the model's pixel-level perceptual alignment while avoiding model collapse. Afterward, to optimize the sensitivity of VLMs' textual and visual embeddings to RS semantic information, bias-guided image–text collaborative optimization is presented to achieve a bilateral interaction of semantic information with the guidance of RS scenes' Bias. Finally, an improved upsampling decoder is employed to obtain the progressive refinement and calibration of the cost map through the integration of multiscale information and textual embeddings. Extensive experiments demonstrate that our method achieves state-of-the-art performance on widely used challenging benchmarks. Code is available at https://github.com/H1NATA111/SGSeg.git**

***Index Terms*—Fine-tune, open-vocabulary segmentation, remote sensing (RS), semantic segmentation.**

I. Introduction

SEMANTIC segmentation of remote sensing (RS) images is one of the most widely researched topics in both RS and computer vision, which has achieved fruitful successes with the advancements of deep learning [1], [2], [3], [4]. However, as illustrated in the upper left portion of Fig. 1, existing semantic segmentation models are limited by a closed vocabulary established during the training phase, which is much smaller than the number of categories focused on in real-world RS scenes [5], [6], [7]. Consequently, when encountering categories absent from the predefined vocabulary in the testing phase, the only solution is to collect additional data, manually annotate them, and retrain the model from scratch. This costly and time-consuming process, combined with the scarcity of RS data, has become an acknowledged bottleneck in the RS community.

Recently, vision-language models (VLMs) [8], [9], such as contrastive language-image pretraining (CLIP) [10], learn aligned representations from billions of image–text pairs and show remarkable generalization capability. As shown in the lower left portion of Fig. 1, this capability has inspired extensive research into open-vocabulary semantic segmentation (OVSS) [5], expanding the vocabulary to a scale that closely resembles that of the real world. The right portion of Fig. 1 illustrates that current mainstream methods, such as CAT-Seg [11], follow the "fine-tune" paradigm. This paradigm involves fine-tuning the visual encoder to capture refined feature representations, which are then correlated with text embeddings to predict pixel-level results [12], [13]. However, despite being effective in the natural scenes, we have found that this direct fine-tune approach performs poorly in the RS scenes.

To investigate this problem further, we try to use a simple open-vocabulary segmenter [14] to predict segmentation results on RS images containing both seen and unseen classes. After fine-tuning the visual encoder, the accuracy of the most seen classes improves, and the accuracy of unseen classes decreases. It is reasonable since CLIP is pretrained on image-level data from natural scenes. The discernible gap of semantic information and data scale between different downstream datasets will lead to noise, causing the visual feature space to collapse in the seen categories [12].

Moreover, we also observed that the accuracy of some classes is notably low, regardless of the application of the fine-tune paradigm. This indicates that CLIP is highly insensitive and inaccurate in perceiving RS semantic concepts [7], as similar textual descriptions can convey diverse and more complex semantic information in RS compared to natural scenes [15], [16]. For example, in natural scenes, a "tree" is usually characterized by its trunk and foliage and is generally perceived as a "large" object. However, from an RS point

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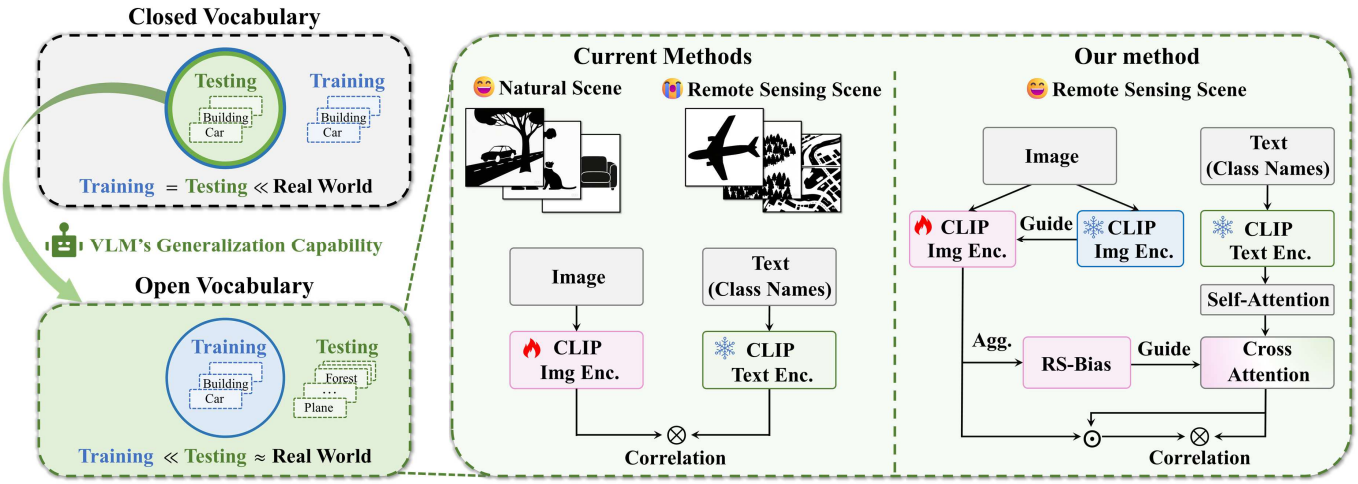


Fig. 1. (Left) Differences between closed vocabulary (traditional semantic segmentation) and open vocabulary (OVSS). (Right) Different frameworks for OVSS. The current "fine-tune" models show insensitivity in perceiving RS semantic cues. Our method softly adapts VLMs to the downstream task of RS semantic segmentation by GC strategy and Bias-CO.

of view, these discriminative characteristics may be absent or even the opposite. Therefore, relying solely on unilateral fine-tuning of the visual encoder is insufficient. It is equally crucial to adapt the text encoder to RS scenes.

Based on the above observations, we consider that an expected open-vocabulary RS semantic segmentation method should: 1) retain the original generalization capability while enhancing the model's pixel-level perception by fine-tuning and 2) achieve bilateral collaborative optimization of visual and text embedding spaces toward perceptual sensitivity of RS scenes. To address this, as opposed to existing methods that entirely rely on direct fine-tuning to achieve adaptations for downstream tasks and specific scenes, we propose a soft-guided open-vocabulary RS semantic segmentation framework. Our proposed framework comprises a generalization compensation (GC) strategy, a bias-guided image–text collaborative optimization (Bias-CO), and an improved upsampling decoder. The main idea is illustrated in the right portion of Fig. 1.

The GC strategy aims to ensure that the visual encoder can enhance pixel-level perception from downstream data while maintaining a strong foundation of generalization capability. We incorporate an additional visual encoder with frozen parameters, which provides soft guidance to transfer the generalization capability of CLIP throughout the fine-tuning process. This transfer is achieved by aligning the representations of the frozen encoder and unfrozen encoder. Specifically, we first employ low-frequency enhancement and average pooling to distill high-level semantic representations, which are believed to be associated with generalization capability [17], [18], [19]. After that, self-distillation loss [20], [21], [22] is applied to align these representations. This alignment not only facilitates the transfer of the inherent generalization capability but also promotes a more robust fine-tuning process.

The Bias-CO aims to tackle CLIP's insensitivity in perceiving RS semantic information. This issue has not been thoroughly explored in current methods [7], [11], [14]. A direct approach is to additionally fine-tune CLIP's text encoder [13]. Unfortunately, this approach leads to the complete collapse of the model. Subsequently, inspired by prior work [23], we add an attention layer after the text encoder and perform cross-attention with the image feature to achieve image–text modality alignment. We initially envision that this operation can be helpful to transfer well-structured visual semantic information into the text embedding. Nevertheless, this widely proven solution fails in the RS task. We attribute this problem to CLIP's simplistic modeling of the text embedding space. CLIP employs a computationally intensive image encoder to extract semantic features from pixel data. In contrast, it utilizes a moderately sized, efficient text encoder to process textual information (with approximately 2.5 times fewer parameters and roughly 50 times less computational complexity) [10], [24], [25]. This asymmetric architectural design implies a fragile alignment between image and text feature spaces. As the image encoder learns semantic information related to novel RS scenes, the visual representations inevitably contain substantial noise and irrelevant information. Direct cross-attention can cause the simplistic text encoder to be disturbed by these noises, rendering it unable to adapt to the complex and changing semantic information in the RS scene, leading to the collapse of the alignment relation.

To tackle this dilemma, we introduce Bias-CO. Specifically, we first design a memory module [26], [27] to dynamically model the visual distribution of the fine-tuned encoder. This distribution is then mapped into a consistent high-dimensional space to obtain an RS scenes Bias (RS-Bias). As the memory module updates, the RS-Bias more accurately captures the semantic representations of RS scenes, thus softly guiding image–text embedding space toward RS scenes. As opposed to current methods, where they utilize point-to-point cross-attention in modality alignment, our bias-guided collaborative optimization enables the implicit transfer of semantic information between different modalities from the perspective of the scene level. RS-Bias can be regarded as a semantic guidance, filtering out noise and redundant information of visual features and retaining only the key semantic information relevant to

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the RS scene. This facilitates the learning process for the text encoder, enabling it to achieve image–text alignment more effectively, thereby preventing the collapse of the embedding space. Moreover, dynamic update of RS-Bias is performed only during the training phase, thus minimizing computational complexity.

In addition, *an improved upsampling decoder* is proposed to gradually refine and calibrate the cost maps. Semantic segmentation greatly benefits from high-resolution feature maps [14]. However, the raw cost maps generated by correlation calculation have a relatively low resolution and high noise [11], [28]. Consequently, directly upsampling the cost map to generate high-quality segmentation results is difficult to tackle the challenges of RS scenes, such as diverse geographical landscapes and varied objects [29], [30], [31]. To address this issue, we incorporate multiple interactions between optimized visual and textual features in the upsampling decoder to mitigate noise and enrich the semantic information of both modalities.

We compare our framework with current state-of-the-art models on iSAID [32], DLRSD [33], LoveDa [34], Vaihingen [35], Potsdam [36], and GaoFen-2 [37] datasets, achieving superior performance and effectively demonstrating the efficacy of our approach. Finally, the contributions of this article are summarized as follows.

1) We propose a soft-guided open-vocabulary RS semantic segmentation framework. This is the earliest work to explore how to softly adapt CLIP to the downstream task of RS segmentation.
2) We design a GC strategy, which utilizes a frozen encoder to implicitly guide the fine-tuning of the unfrozen encoder, thus enhancing the model's pixel-level perception while avoiding catastrophic forgetting.
3) To optimize the sensitivity of CLIP's textual and visual embeddings to RS semantic information, Bias-CO is presented to achieve a bilateral interaction of semantic information with the guidance of RS-Bias.
4) We propose an improved upsampling decoder that achieves the progressive refinement and calibration of the cost map through the integration of multiscale features and textual embeddings.

The remainder of the article is organized as follows. Section II presents related work. In Section III, we introduce our proposed method. Then in Section IV, we demonstrate and analyze the method with experimental results. Finally, Section V concludes the article.

II. Related Work

A. Vision-Language Models

Recently, the emergence of large-scale VLMs [8], [9] has energized the field of computer vision. Among these, CLIP [10] stands out as a pioneering framework that learns joint embeddings through contrastive training on 400 million image–text pairs. By aligning visual and textual semantics in a shared latent space, CLIP demonstrates remarkable zero-shot transferability. Building upon CLIP, subsequent works like CLIP-ConvNeXt [38], [39] enhance architectural compatibility by replacing the original vision transformer (ViT) [40] backbone with a hierarchical ConvNeXt [38] structure. This modification improves computational efficiency and spatial feature preservation through its convolutional inductive bias, making it more suitable for dense prediction scenarios.

B. Open-Vocabulary Remote Sensing Semantic Segmentation

Traditional semantic segmentation methods mainly contain FCN-based approaches [1] and transformer-based approaches [41], [42], [43]. However, these models are limited by a closed vocabulary. To address this, ZegFormer [44] and ZSseg [45] proposed two-stage frameworks for dealing with the task. Typically, they first learn to predict class-agnostic region proposals similar to [46], and feed them to CLIP for final predictions. MaskCLIP [47] leverages the self-attention map from CLIP to refine the region proposals. MAFT [12] and MAFT+ [13] find that fine-tuning the CLIP's encoder can enhance the model's performance for downstream tasks and scenes. Subsequent models have adopted a similar fine-tune paradigm. Recently, CAT-Seg [11] and SED [14] proposed one-stage costmap-based frameworks, effectively adapting CLIP to the downstream task of segmentation by fine-tuning its encoders. However, these methods acknowledge that the fine-tune paradigm may lead to a loss of generalization capability [12], [14]. This loss negatively impacts the performance of these methods designed for natural scenes when applied to the open-vocabulary RS semantic segmentation task.

More recently, the first open-vocabulary RS semantic segmentation framework OVRS [7] integrates rotational scale features with encoder features to address the challenges posed by rapidly changing orientations and significant scale variations in RS scenes. However, these handcrafted features are neither accurate nor stable, and OVRS does not fundamentally address the issue of generalization capability loss.

Following OVRS [7], SegEarth-OV [48] proposed to achieve open-vocabulary RS semantic segmentation with a training-free approach. However, the VLMs are pretrained on natural scenes; this "training-free" paradigm lacks the integration of RS knowledge, which hinders the model's ability to fully perceive the complex semantic information in RS scenes. As a result, its generalization capability across various image domains is limited, leading to inconsistent segmentation performance.

III. Method

In this section, we introduce our soft-guided OVSS framework for RS images, named SGSeg. Then, we describe the details of the proposed GC strategy, Bias-CO, and upsampling decoder.

Problem Setting: Open-vocabulary RS semantic segmentation addresses the task of training a segmentation model capable of segmenting novel categories using text descriptions. Given a RS image and a set of candidate categories $C = \{C_n\}$ for $n = 1, 2, \ldots, N_s, \ldots, N$, where C_n denotes textual description based on nth category, N_s is the number of seen categories and N is the number of categories, OVSS assigns a

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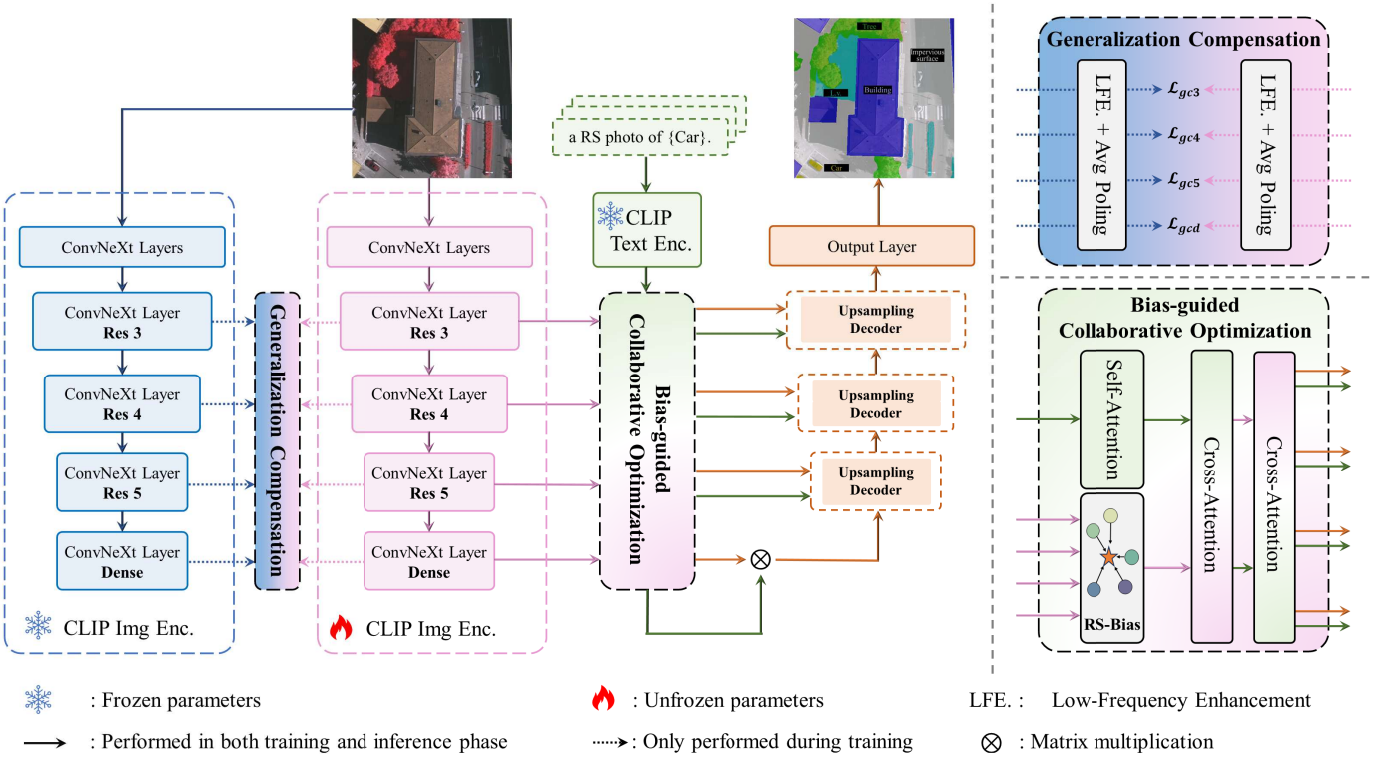


Fig. 2. *Overview of the proposed soft-guided open-vocabulary RS semantic segmentation framework.* Our VLM employs a pretrained CLIP with ConvNeXt backbone. First, we implement the GC strategy to guide the fine-tuning of the visual encoder. Subsequently, we implement Bias-CO to align visual and textual embeddings and generate cost maps. Finally, we design an improved upsampling decoder to progressively refine and calibrate the cost maps.

class label for each pixel in image I. Different from classical semantic segmentation tasks, open-vocabulary segmentation is additionally challenged by varying I, given as free-form text description C_n.

A. Overview

As shown in Fig. 2, given an RGB image I and a set of categories C, we first utilize a visual encoder to extract image features. In contrast to the conventional paradigm, our image encoder comprises two subencoders, CLIP_H and CLIP_C. The subencoders are initialized from the visual branch of a pretrained CLIP. Specifically, the parameters of CLIP_H are unfrozen and fine-tuned during training, while CLIP_C remains frozen to maintain CLIP's original generalization capability. The subencoders each produce multiscale feature representations of the input image I. We donate these representations as sets $F_H = \{F_H^i \mid i \in 3, 4, 5, \text{dense}\}$ and $F_C = \{F_C^i \mid i \in 3, 4, 5, \text{dense}\}$, where i indicates the stage of progressive hierarchical downsampling. The strides of F_H^i and F_C^i with respect to the input resolution are 4, 8, 16, and 16 pixels, respectively. When i = dense, F_C^{dense} corresponds to the value from the last attention layer of CLIP's image encoder. This feature has been proven to be dense patch-level, demonstrating intrinsic potential for dense prediction, particularly in semantic segmentation.

During the fine-tuning process, F_H is progressively refined into pixel-level feature representations. GC strategy is designed to utilize F_C to supervise and guide this process. This approach helps prevent F_H from overfitting to seen categories during training. Details on GC strategy are provided in Section III-B.

Simultaneously, based on the C, prompt templates are employed to generate various textual descriptions T, for example, "a photo of a C_n." The size of T is $N \times L$, where N represents the total number of categories and L denotes the number of templates per category. These textual descriptions are then fed into the CLIP's text encoder to obtain text embeddings E, which have a size of $N \times L \times D$, where D denotes the number of channels. Subsequently, F_H^i from the unfrozen visual encoder and the text embeddings E are fed to the Bias-CO. Details on this step are provided in Section III-C. By calculating the correlation between the aligned text embeddings $\bar{E}$ and the dense visual features $\bar{F}_H^{\text{dense}}$, we obtain a pixel-level cost map M

$$M(i, j, n, l) = \frac{\bar{F}_H^{\text{dense}}(i, j) \cdot \bar{E}(n, l)}{\left\|\bar{F}_H^{\text{dense}}(i, j)\right\| \left\|\bar{E}(n, l)\right\|} \tag{1}$$

where i and j donate the 2-D spatial position, n represents the index of text embeddings, and l denotes the index of templates. The size of the cost map M is $N \times H/16 \times W/16$. The cost map M can be interpreted as calculating the correlation slice for each category to coarse, low-resolution predictions. However, these coarse predictions still contain significant noise and misclassifications. Therefore, we introduce an improved upsampling decoder to leverage the optimized multiscale visual features $\bar{F}^i$ and text embeddings $\bar{E}$ as sources of spatial and class-discriminative information, which iteratively refines

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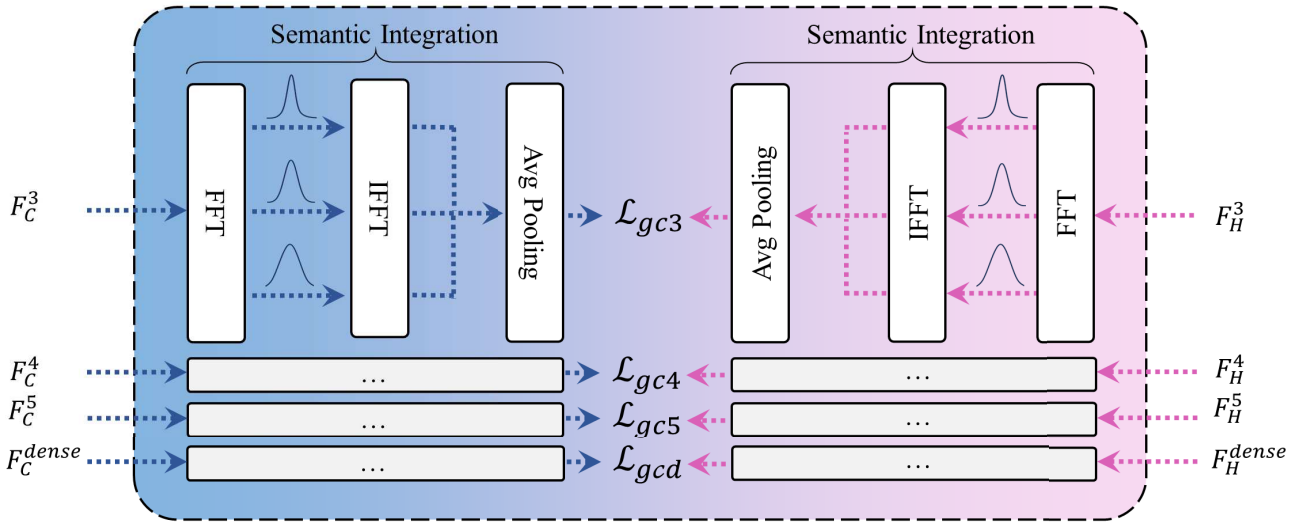


Fig. 3. *Detailed illustration of our GC strategy.* We align different stages of frozen and unfrozen visual features to mitigate the loss of generalization capability.

the cost map. The details of upsampling decoder are described in Section III-D.

B. Generalization Compensation

GC strategy aims to mitigate the loss of generalization capability due to fine-tuning. As illustrated in Fig. 3, during the training phase, we utilize a frozen encoder to guide and regulate the unfrozen encoder. This is achieved by aligning the high-dimensional semantic space that represents generalization capability, thereby preventing the model from collapsing to seen categories. However, direct per-pixel alignment is not feasible, as the associated local details may bring harmful texture noise, which can disturb the perceptual refinement of the unfrozen encoder and lead to ineffective fine-tuning. To address this issue, we designed a semantic integration network composed of multiple low-frequency enhancements and global pooling layers, enabling the consistency of integrated features and achieving alignment in the high-dimensional semantic space.

Some theoretical research on spectral domain neural networks [46], [49], [50] indicates that low-frequency components correspond to high-level semantic information and are generally considered to be related to the generalization capability of the model [17], [18], [19]. Inspired by this, we propose to utilize low-frequency enhancement to suppress potential noise and capture high-level semantic information. As shown in Fig. 3, take F_C^3 as an example, we first define low-pass filters

$$g_\alpha^3 = \text{Gaussian}\left(\frac{H}{16}, \frac{W}{16}, \sigma_\alpha\right) \tag{2}$$

where g_α^3 denotes the filter coefficient maps corresponding to F_C^3, with both sharing sizes of $H/16 \times W/16$. The coefficient map has a value of 0 at the center and gradually increases following a Gaussian profile as the distance from the center increases. α denotes various cutoff frequencies, enabling different degrees of low-frequency filtering of features. Subsequently, we perform the Fourier transform to F_C^3

$$f_C^3 = \text{FFT}\left(F_C^3\right) \odot g_\alpha^3. \tag{3}$$

Due to the absence of frequency centering, the center region of the F_C^3 after FFT corresponds to high frequencies, while the surrounding region represents low frequencies. We then perform a Hadamard product ($\odot$) between F_C^3 and g_α^3 to implement high-frequency filtering, resulting in the low-frequency-enhanced spectral feature f_C^3. Finally, we perform an inverse Fourier transform to f_C^3

$$F_C'^3 = \text{IFFT}\left(f_C^3\right). \tag{4}$$

After low-frequency enhancement, $F_C'^3$ is fed to multiscale average pooling layers to further suppress noise and irrelevant details

$$P_C^3 = ML - \text{AvgPool}\left(F_C'^3,\ \lambda\right) \tag{5}$$

where λ is a manually defined hyperparameter (empirically set to 1, 2, or 4) that determines the scale of the average pooling layer; when $\lambda_i = 1$, it denotes global average pooling. Specifically, the average pooling operation at each scale generates a high-level semantic representation with size of $\lambda_i \times \lambda_i$. We then repeat the smaller-scale representations to match the largest scale and concatenate them for subsequent loss computation.

Finally, we apply the same procedure to F_H^3 to obtain its corresponding high-level semantic representation P_H^3 and align them with self-distillation loss

$$\text{Loss}_{gc}\left(P_C^3, P_H^3\right) = \begin{cases} 0.5 \cdot \left(P_H^3 - P_C^3\right)^2, & \text{if } \left|P_H^3 - P_C^3\right| < 1 \\ \left|P_H^3 - P_C^3\right| - 0.5, & \text{otherwise.} \end{cases} \tag{6}$$

We repeat the above operations to align the features of the visual encoder at $4, 5,$ dense stages. These alignments ensure that the visual encoder benefits from downstream data while preserving its robust generalization capability and avoiding catastrophic forgetting.

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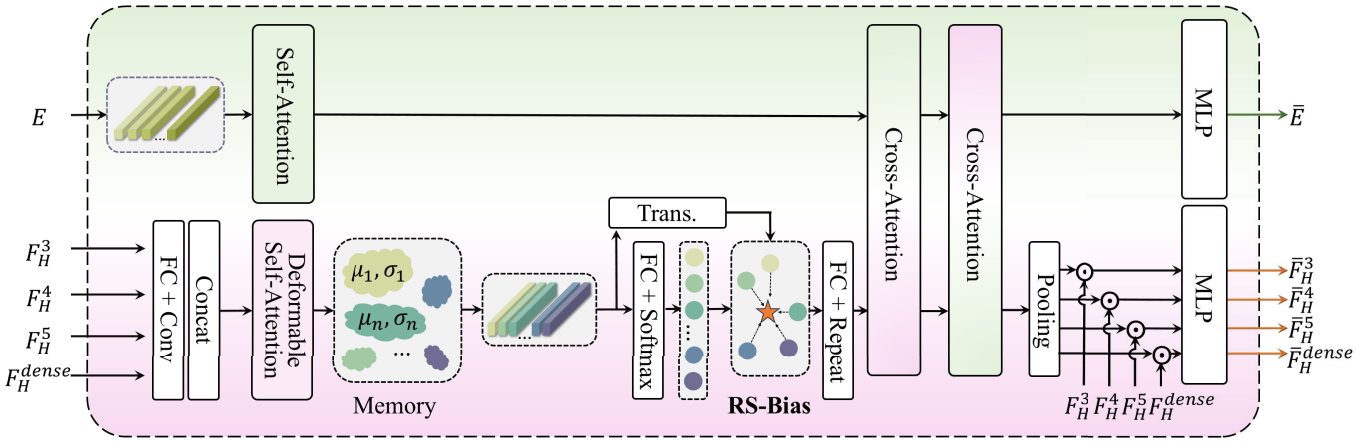


Fig. 4. *Detailed illustration of our Bias-CO.* We first aggregate the RS-Bias from visual features and then utilize it to guide the interaction between visual and textual embeddings.

C. Bias-Guided Image–Text Collaborative Optimization

The robust image–text alignment capability of VLMs is crucial for enabling semantic segmentation models to overcome the limitation of closed vocabulary. However, the great gap between the two modalities has emerged as a recognized bottleneck. In the RS scene, this issue is further exacerbated because of the limited sensitivity of the pretrained text encoder to RS-specific semantic information. These double gaps make existing methods struggle to tackle the unique challenges in RS (RS). To address these challenges, we propose a Bias-CO. Instead of enforcing a hard alignment between the modalities, we capture scene-level semantic representation RS-Bias to softly guide the cross-modality knowledge transformation, ultimately achieving collaborative optimization.

As illustrated in Fig. 4, given multiscale visual feature $F_H = \{F_H^i \mid i \in 3, 4, 5, \text{dense}\}$ and text embedding E. The fine-tuning process and GC strategy have enabled the visual encoder to capture well-structured RS-related semantic information. However, direct cross-attention not only fails to transfer the semantic information to the text embedding but may also lead to the collapse of the text encoder due to its comparatively simpler modeling. Therefore, we propose to reconstruct the visual feature to integrate multiscale semantic information

$$\begin{aligned} F &= \text{Conv}^3\left(F_H^3\right) \oplus \text{Conv}^4\left(F_H^4\right) \\ &\oplus \text{Conv}^5\left(F_H^5\right) \oplus \text{Conv}^{\text{dense}}\left(F_H^{\text{dense}}\right) \end{aligned} \tag{7}$$

where Conv^i represents convolutional layers that reconstruct the multiscale visual features to the same resolution, $\oplus$ denotes channel concatenation, the size of F is $H/16 \times W/16 \times D$. We then feed the reconstructed feature F into deformable self-attention to further enhance it

$$\text{DSA}(F) = \text{softmax}\left(\frac{Q(F) \otimes K^T(F)}{\sqrt{d_k}} + B\right) \otimes V(F) \tag{8}$$

where $Q(\cdot)$, $K(\cdot)$, and $V(\cdot)$ denote the linear projections that produce the query, key, and value. d_k is the dimension of key. B is the relative position bias introduced, and $\otimes$ represents matrix multiplication. For convenience, we label the enhanced feature as F'. Then, we employ a simple memory module to cluster F'

$$R_n = \left\{ F'_{[i,:]}{}^{\frac{\text{HW}}{64} \times C} \mid (GT_i = n) \cap (i \subseteq \text{HW}) \right\} \tag{9}$$

where R_n represents the class-level feature for each seen class, with size of $N_s \times D$, $1 \le n \le N_s$, N_s denotes the number of seen class. The index i indicates the ith pixel in matrix F', and GT denotes the category label of ground truth.

1) Update of RS-Bias: As the memory module updates, RS-Bias can more accurately capture the semantic representations of RS scenes. This update is only performed during the training phase and is shared across datasets during the inference phase. Specifically, inspired by [26], we store and update only two hyperparameters, mean and variance, for each class-level semantic representation. Consequently, the memory module's storage is only a $N_s \times 2$ matrix, where the mean and variance are initialized to 0 and 1. This strategy significantly reduces the computational complexity. The update strategy of mean and variance is as follows:

$$\begin{aligned} \mu_n &= \beta \cdot \mu_{n-1} + (1-\beta) \cdot \text{Mean}(R_n) \\ \sigma_n &= \beta \cdot \sigma_{n-1} + (1-\beta) \cdot \text{Std}(R_n) \end{aligned} \tag{10}$$

where ω denotes the current number of iterations, μ and σ are the mean and standard deviation, and β is an artificial hyperparameter (empirically set to 0.1), preventing outliers in individual samples that cause the summary distribution to deviate significantly from the true distribution. This class-level semantic aggregation effectively simplifies the complex modeling of visual features and reduces the risk of text encoder collapse during subsequent modality alignment. However, such operation may lead to overfitting to the seen classes. To address this, we further aggregate class-level representation to model a universally applicable scene-level distribution. Specifically, we first employ a fully connected layer to compress the category-level semantic representations into $N_s \times 1$, thereby integrating the semantic information and generating the class embeddings ε

$$\varepsilon = \text{softmax}(\text{FC}(R_n)) \tag{11}$$

where the size of ε is $N_s \times 1$. Softmax is applied to constrain the value between 0 and 1. The class embeddings donate the

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global semantic information for each class. Fundamentally, the RS semantic space is constituted by these prototypical classes. Therefore, by leveraging the shared semantic information among these representative classes, we can derive the universal representation of the RS semantic space, i.e., scene-level semantic representation RS-Bias

$$\text{RS} - \text{Bias} = \varepsilon \otimes \text{Transpose}\left(R_n\right) \tag{12}$$

where RS-Bias can be conceptualized as a beneficial scene data bias. This progressive modeling of the semantic space, transitioning from category-level to scene-level, emphasizes critical universal knowledge while effectively suppressing class-specific noise.

Simultaneously, directly fine-tuning the text encoder can result in the collapse of the method. Consequently, we incorporate a fully connected layer and a self-attention layer following the text encoder to refine the text embeddings

$$E' = \text{SA}\left(\text{FC}\left(E\right)\right) \tag{13}$$

where SA denotes the self-attention, and the size of E' is $N_s \times D$. After that, we adjust the RS-bias by a fully connected layer and repeat it N_s times to match the dimensions of the text embeddings, the output is labeled as RS-Bias', which also has a size of $N_s \times D$.

Subsequently, we utilize RS-Bias' to softly guide the alignment between the text embeddings and visual features. To facilitate the transfer of semantic information, we add an image-to-text cross-attention and a text-to-image cross-attention for feature fusion for infusing scene-level semantic knowledge into the text embeddings and feeding back to optimize RS-Bias'. This process can be formulated as

$$\begin{aligned} &CA\left(E', \text{RS-Bias}'\right) \\ &= \text{softmax}\left(\frac{Q\left(E'\right) \otimes K^T\left(\text{RS} - \text{Bias}'\right)}{\sqrt{d_k}}\right) \\ &\quad \otimes V\left(\text{RS} - \text{Bias}'\right) \end{aligned} \tag{14}$$

$$\begin{aligned} &CA\left(\text{RS} - \text{Bias}'', E''\right) \\ &= \text{softmax}\left(\frac{Q\left(\text{RS} - \text{Bias}''\right) \otimes K^T\left(E''\right)}{\sqrt{d_k}}\right) \\ &\quad \otimes V\left(E''\right) \end{aligned} \tag{15}$$

where RS-Bias" and E'' represent the intermediate features from two stages of cross-attention. For convenience, we donate the final outputs as RS-$\dot{\text{B}}$ias and $\dot{E}$. This process effectively achieves a bidirectional semantic transfer because RS-$\dot{\text{B}}$ias also captures valuable information from the text embedding. After that, we feedback RS-$\dot{\text{B}}$ias into the multiscale visual features

$$\bar{F}_H^i = \text{MLP}\left(P\left(\text{RS} - \text{Bias}\right) \odot F_H^i\right) \tag{16}$$

where P denotes average pooling, and the MLP consists of two fully connected layers designed to enhance learning capacity. Finally, $\dot{E}$ is similarly fed into an MLP layer to obtain the optimized text embedding, donated as $\bar{E}$.

Compared to point-to-point feature alignment, the semantic complexity of RS-Bias is similar to that of the text embeddings; as a result, it is "soft." This soft-guided approach mitigates the risk of model collapse due to the double gaps of modality and scene discrepancies, thereby achieving effective image–text collaborative optimization.

D. Upsampling Decoder

Semantic segmentation greatly benefits from high-resolution feature maps [1], [3]. However, the raw cost maps M generated by correlation calculation have a relatively low resolution and high noise. Consequently, directly upsampling the cost map to generate high-quality segmentation results is not beneficial. To address this issue, we propose an improved upsampling decoder, as illustrated in Fig. 5. We incorporate optimized text embeddings and multiscale visual features to enable the gradual refinement and calibration of the cost map M.

Research on VLMs [23] indicates that more interaction between visual and textual features in the pipeline enables the model to perform better. Inspired by this, we perform linear attention [51] at each layer of the decoder. Specifically, we first project M to the channel of $D/4$, labeled as M_1. The design enables the decoder to be lightweight. Then, we utilize a fully connected layer to project the text embeddings to the same size as M_1 and then repeat them

$$\bar{E}' = \text{Repeat}\left(\text{FC}\left(\bar{E}\right)\right) \tag{17}$$

where the size of $\bar{E}'$ is $N \times D/4 \times H/16 \times W/16$. The embedding $\bar{E}'$ is then concatenated to the cost map M_1 before query and key projections in the linear attention layer

$$M_1' = \text{LA}\left(Q\left(\bar{E}' \oplus M_1\right),\ K\left(\bar{E}' \oplus M_1\right), V\left(M_1\right)\right) \tag{18}$$

where LA denotes linear attention, following [51]. This approach emphasizes significant features, highlighting the correlation of each cost map slice to its corresponding category, thereby correcting early misclassification caused by noise.

Some research about RS images processing indicates that multiscale features effectively capture semantic information on different scales, complex textures, and contexts [3]. Inspired by this, we extend our focus beyond deep features to include shallow features during the decoding phase. This design can enhance the local details of the cost map. As illustrated in Fig. 5, we first employ a convolutional layer to adjust the channel of the shallow feature. Then, the normalization and repeat operations are performed. Taking the first decoder as an example, this process can be expressed as follows:

$$\overline{F'}_H^5 = \text{Repeat}\left(\text{LN}\left(\text{Conv}\left(\bar{F}_H^5\right)\right)\right) \tag{19}$$

where LN is layer normalization, and Conv denotes a 1×1 convolutional layer. The size of $\overline{F'}_H^5$ is $N \times D/4 \times H/16 \times W/16$, same as cost map. The feature $\overline{F'}_H^5$ is then concatenated to the cost map M'_1 before query and key projections in the cross attention layer to achieve detail refinement

$$M_1'' = \text{CA}\left(Q\left(\overline{F'}_H^5 \oplus M_1'\right),\ K\left(\overline{F'}_H^5 \oplus M_1'\right), V\left(M_1'\right)\right). \tag{20}$$

Similarly, CA denotes the cross attention, which is employed to enrich the cost map with detailed information from shallow features. This approach enhances the model's ability to capture both local and global information in complex scenes, thereby

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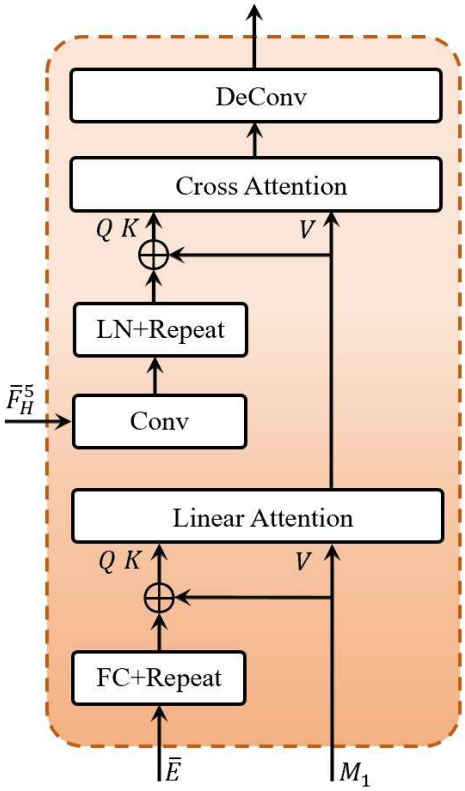


Fig. 5. *Detailed illustration of our improved upsampling decoder.* This process is repeated three times during the upsampling phase (the illustration here represents the first process) to perform as much modality interaction and multiscale feature fusion as possible.

improving segmentation accuracy. Finally, we feed M_1'' into a learnable transposed convolution layer to perform upsampling

$$M_2 = \text{DeConv}\left(M_1''\right) \tag{21}$$

where the size of M_2 is $N \times D/4 \times H/8 \times W/8$.

After that, we repeat the above operations to progressively refine and calibrate the cost map while increasing the resolution. Finally, the output from the last decoder is fed to the segmentation head to obtain fine-grained pixel-level segmentation labels.

IV. Experiments

Following existing open-vocabulary RS semantic segmentation methods [7], we use the large-scale dataset iSAID [32] or DLRSD [33] to train the model. With the trained model, we conduct experiments on multiple widely-used semantic segmentation datasets (Vaihingen [35], Potsdam [35], GaoFen-2 [37], and LoveDA [34]) to demonstrate the effectiveness of the proposed method and compare it with state-of-the-art methods in the literature.

A. Dataset Description

To comprehensively evaluate the proposed method, we conducted our experiments on widely used remote sensing semantic segmentation datasets, including iSAID, DLRSD, Vaihingen, GaoFen-2 and LoveDA. The processing methods and detailed descriptions of these datasets are as follows:

1) *Instance Segmentation in Aerial Images Dataset (iSAID):* The iSAID [32] was modified from a large-scale object detection dataset, DOTA. This large-scale and densely annotated dataset contains 655 451 object instances for 15 categories across 2806 high-resolution images. The labeling categories include 15 landcover classes: 1) ship; 2) storage tank; 3) baseball diamond; 4) tennis court; 5) basketball court; 6) ground track field; 7) bridge; 8) large vehicle; 9) small vehicle; 10) helicopter; 11) swimming pool; 12) roundabout; 13) soccer ball field; 14) plane; and 15) harbor. We adopt the same dataset division as [26] and [29].
2) *DLRSD Dataset:* DLRSD [33] is a dense labeling dataset that can be used for semantic segmentation. The average image size is 256×256. The DLRSD is split into 5601 training and 1401 validation images with annotation containing 16 categories: 1) airplane; 2) bare soil; 3) buildings; 4) cars; 5) chaparral court; 6) dock; 7) field; 8) grass; 9) mobile home; 10) pavement; 11) sand; 12) sea; 13) ship; 14) tanks; 15) trees; and 16) water. The dataset division is following [7].
3) *Vaihingen Dataset:* The ISPRS 2-D semantic labeling Vaihingen dataset [35] is provided by the International Society for Photogrammetry and RS, which consists of 33 high-resolution true orthophoto tiles and corresponding digital surface models, and the DSM band channel is abandoned in our experiment. The average image size is 2494×2064. We adopt the same dataset division as [26] and [29]. The categories of the labeling include six landcover classes: 1) impervious surface; 2) building; 3) low vegetation; 4) tree; 5) car; and 6) clutter.
4) *GaoFen-2 Dataset:* GaoFen-2 dataset [37] is a more challenging dataset compared with the ISPRS 2-D Semantic Labeling Vaihingen data. It contains 500 satellite images collected from the GaoFen-2 satellite over different geographic locations in China, which is featured by high intraclass diversity and low interclass separability. The dataset division is following [26] and [30]. The labeling categories include nine categories: 1) road; 2) building; 3) tree; 4) grass; 5) bare land; 6) water; 7) transportation; 8) impervious surfaces; and 9) others.
5) *LoveDA Dataset:* LoveDA dataset [34] consists of 5987 high-resolution images of 1024×1024 pixels and 30 cm in spatial resolution. The data include 18 complex urban and rural scenes and 166,768 annotated objects from three different cities (Nanjing, Changzhou, and Wuhan) in China. We select 1669 images for validation following [52]. The categories of the labeling include eight landcover classes: 1) no-data; 2) background; 3) building; 4) road; 5) water; 6) barren; 7) forest; and 8) agriculture. We perform experiments on these categories other than "no-data."

B. Evaluation Metrics

The mean intersection over union (mIoU) is chosen for the evaluation accuracy, which is calculated as follows:

$$\text{IoU}_i = \frac{x_{ii}}{\sum_{j=1}^{n} x_{ij} + \sum_{j=1}^{n} x_{ji} - x_{ii}} \tag{22}$$

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TABLE I

COMPONENTWISE ABLATIONS ON iSAID. WE CONDUCT AN ABLATION STUDY BY GRADUALLY ADDING COMPONENTS TO THE BASELINE. THE BEST-PERFORMING RESULTS ARE PRESENTED IN BOLD

Baseline	GC	Bias-CO	Decoder	DLRSD	Vaihingen	Potsdam	LoveDA	GaoFen-2
✓				24.37	16.13	19.9	14.65	19.46
✓	✓			25.59	22.71	23.57	19.96	20.33
✓	✓	✓		27.25	24.08	26.84	20.07	22.98
✓	✓	✓	✓	**28.52**	**26.76**	**29.26**	**21.97**	**23.85**

$$\text{mIoU} = \frac{1}{n}\sum_{i=1}^{n} \text{IoU}_i \tag{23}$$

where x_{ij} represents the number of instances of class i predicted as class j, and n is the number of classes.

For the model efficiency, the computational complexity and inference speed are evaluated, i.e., the number of model parameters and frames per second.

C. Training Settings

We adopt the pretrained vision-language model CLIP as the base model, where the backbone ConvNeXt-L is used as the visual encoder. The same prompt templates are used for each dataset, with a total of 80. The exact prompt templates follow OVRS [7] and CAT-Seg [11]. During training, the AdamW optimizer is adopted with betas of (0.9, 0.999) and a weight decay of 0.01. The initial learning rate is set to 0.00006, and a "Poly" schedule with power 0.9 is applied. The model is trained for a total of 60 000 iterations on the training dataset, and the batch size is set to 4. All the networks are implemented using 1 RTX A6000 GPU with PyTorch and Detectron2 frameworks.

D. Analysis and Ablation Study

1) Componentwise Ablations: In this section, we have performed an ablation analysis of the different components to evaluate the effectiveness and generalization capability of our method. We remove the GC strategy, Bias-CO, and replace our proposed upsampling decoder (decoder) with a simple multilayer deconvolution as the baseline model. As shown in Table I, we start with the baseline model and gradually add each design. We selected the iSAID dataset as the training dataset, while the other datasets are used for validation. This design is made because, although iSAID includes many classic RS target categories, the majority of these categories are not completely the same as those in the other datasets. The presence of numerous unseen classes makes experiments under this design particularly challenging, thereby facilitating the assessment of the model's generalization capability.

1) *Baseline:* The direct fine-tuning paradigm leads to catastrophic forgetting in the model, resulting in a loss of essential generalization capability and consequently poor performance. The baseline only obtains the mIoU scores of 24.37%, 16.13%, 19.9%, 14.65%, and 19.46% on DLRSD, Vaihingen, Potsdam, LoveDa, and GaoFen-2 (Row 1).

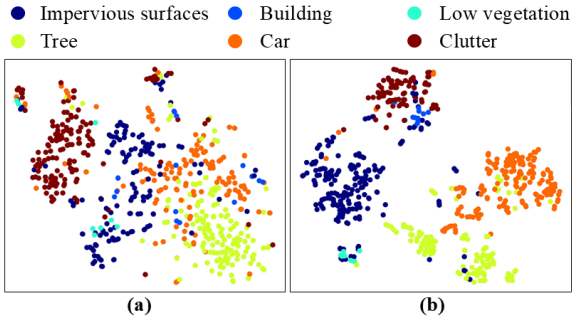


Fig. 6. *t-SNE analysis for the contribution of GC strategy.* We show t-SNE visualization of the dense image embeddings of visual encoders in the Potsdam dataset (trained in iSAID). (a) Baseline. (b) Baseline+GC.

2) *Baseline+GC:* Then, the GC strategy utilizes a frozen encoder to guide and regulate the unfrozen encoder, thereby mitigating the loss of generalization capability. When adding GC strategy, it has a mIoU of 25.59%, 22.71%, 23.57%, 19.96%, and 20.33% on validation datasets, which outperforms the baseline by 1.22%, 6.58%, 3.67%, 5.31%, and 0.87%. Besides, we show the t-SNE visualization of the dense image embeddings of visual encoders in the Potsdam dataset (trained in iSAID). From Fig. 6(a), we can observe that the structure of the clusters is not well-formed for each class. This issue arises because most classes in Potsdam are not seen in iSAID, and direct fine-tuning of the baseline model leads to a collapse toward seen classes. In contrast, we observe well-structured clusters in Fig. 6(b) showing that the GC strategy is helpful for retaining the original generalization capability. This well generalization capability enables our model to accurately model the distributions for both seen and unseen classes, showing the effectiveness of our proposed GC strategy.

3) *Baseline+GC+Bias-CO:* When further integration of Bias-CO, RS-Bias is aggregated to infuse comprehensive RS knowledge into text embeddings and the visual feedback feature. The method with collaborative optimization significantly outperforms the performance of *Baseline+GC* by a mIoU of 1.66%, 1.37%, 3.27%, 0.11%, and 2.65% on validation datasets. This result shows that GC and Bias-CO can work well together to enable visual and textual features to be more effective and reliable, which demonstrates the good reliability and generalization capability of our proposed components.

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TABLE II
ANALYSIS OF COMPUTATIONAL COMPLEXITY AND INFERENCE SPEED

Baseline	GC	Bias-CO	Decoder	Parameters (M)	Frame Per Second (FPS)
✓				436.52	11
✓	✓			436.52	11
✓	✓	✓		457.81	10
✓	✓	✓	✓	461.29	10

TABLE III
COMPARISON BETWEEN VLMS ON iSAID. THE BEST-PERFORMING RESULTS ARE PRESENTED IN BOLD

Model	VLM	DLRSD	Vaihingen	Potsdam	LoveDA	GaoFen-2
SED [14]	ConvNeXt-L [39]	25.45	16.43	18.32	11.6	19.2
CAT-Seg [11]	ViT-L/14 [10]	22.13	14	15.79	19.75	12.71
OVRS [7]	ViT-L/14 [10]	20.67	20.82	19.94	17.43	19.39
Ours	ViT-L/14 [10]	**29.1**	23.14	27.15	20.78	20.15
Ours	**ConvNeXt-L [39]**	28.52	**26.76**	**29.26**	**21.97**	**23.85**
Ours	RS-CLIP [25]	24.57	19.95	24.08	16.43	17.46
Ours	EVA-CLIP [24]	31.33	26.96	30.58	22.01	25.9

4) *Baseline+GC+Bias-CO+Decoder:* The improved upsampling decoder is introduced to fully leverage the optimized text embeddings and multiscale visual features, thereby gradually refining and calibrating the raw cost maps. After integrating an improved upsampling decoder, the segmentation performance is improved by 1.27%, 2.68%, 2.42%, 1.9%, and 0.87% on validation datasets with *Baseline+GC+Bias-CO*. In overall analysis, our proposed *Baseline+GC+Bias-CO+Decoder* network exhibits optimal performance with 28.52%, 26.76%, 29.26%, 21.97%, and 23.85% mIoU, which is significantly higher than the Baseline 4.15%, 10.63%, 9.36%, 7.32%, and 4.39% mIoU. These results well demonstrate the reliability, effectiveness, and generalization capability of our proposed EMIDI.

Furthermore, Table II demonstrates the analysis of complexity comparison and inference speed, including the model parameters and frame per second. These metrics are calculated under the same settings during inference. "Parameters" refers to the number of parameters in the whole complete model (including the vision-language model). "Frame Per Second" refers to the average number of 512×512 input images that the model can segment per second during inference time. All the numbers are obtained on a single NVIDIA RTX A6000 GPU.

As seen, our approach shows no material difference in model complexity or inference speed compared with the baseline method, despite introducing additional components. This advantage is attributable to the decoupled design of the GC strategy and the bias-guided image–text collaborative optimization. Specifically, the additional visual encoder and self-distillation alignment in the GC strategy are not performed at inference time. Therefore, *Baseline* and *Baseline+GC* exhibit the same computational complexity and inference speed, with values of 436.52 M and 11 FPS, respectively. Moreover, the RS-Bias is dynamically updated only during training, compared with *Baseline*, *Baseline+GC+Bias-CO* increases the computational complexity by 21.92 M and reduces inference speed by 1 FPS, which is tolerable. The revised decoder incurs a slight increase in computational complexity due to the introduced multiscale features, with negligible impact on inference speed.

2) Comparison Between VLMs: Table III presents results for methods that use VLMs of different architectures and versions. Overall, our method significantly outperforms all competing methods. These results highlight the superiority of our approach, which remains independent of the performance of pretrained VLMs.

Moreover, we employ these various VLMs within our proposed method to further investigate the impact of different encoder architectures on the semantic segmentation task of the RS scene. Specifically, as shown in Table III *(Lines 1–5)*, ViT-L/14 [40] is a classical Transformer architecture, whereas ConvNeXt-L [38] represents a conventional convolutional neural network (CNN) architecture. The model with the VLM of ConvNeXt-L has a mIoU of 28.52%, 26.76%, 29.26%, 21.97%, and 23.85% mIoU, which is better than the model with ViT-L/14 on Vaihingen, Potsdam, LoveDa, and GaoFen-2. These results demonstrate that the model employing ConvNeXt-L exhibits superior performance, as the CNN structure offers rich local information, thereby enabling the model to more effectively address the challenges in RS scenes like complex scale diversity and segmentation of small samples at extreme foreground and background scales. This result is consistent with established viewpoints in the domain of RS image processing.

In addition, Table III *(Lines 6 and 7)* demonstrates the analysis of some special VLM architectures beyond ConvNeXt-L

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TABLE IV
ANALYSIS ON FINE-TUNING OF CLIP ON iSAID. WE ONLY FINE-TUNE THE IMAGE ENCODER EXCEPT FOR THE LAST ITEM. THE BEST-PERFORMING RESULTS ARE PRESENTED IN BOLD

Model	DLRSD	Vaihingen	Potsdam	LoveDA	GaoFen-2
Without F.T.	15.03	12.11	14.69	14.94	15.93
Attn F.T.	24.48	21.59	26.26	19.66	19.25
QK F.T.	27.39	22.57	25.97	19	18.79
KV F.T.	26.65	**26.89**	29.15	14.82	18.47
Full F.T.	**28.52**	26.76	**29.26**	**21.97**	**23.85**
Full F.T. (Img.+Txt.)	8.34	2.69	6.34	5.7	3.32
Full F.T. (Img.+Txt.CA)	12.27	8.53	8.91	10.07	6.54

TABLE V
HYPERPARAMETER ANALYSIS OF σ_α ON iSAID. THE BEST-PERFORMING RESULTS ARE PRESENTED IN BOLD

σ_α	VLM	DLRSD	Vaihingen	Potsdam	LoveDA	GaoFen-2
$[9, 5, 1]$	ConvNeXt-L	28.03	26.21	28.38	21.59	22.32
$[9, 5, 3]$	ConvNeXt-L	27.65	26.43	28.47	20.96	22.99
$[9, 7, 1]$	ConvNeXt-L	27.89	26.2	28.42	21.65	22.05
[9, 7, 3]	ConvNeXt-L	**28.52**	**26.76**	**29.26**	**21.97**	**23.85**

and ViT-L/14. We attempted to employ the RS-CLIP [25] specifically designed for the RS scene as VLM. However, this modified version of CLIP focuses solely on the global [CLS] token. Even though patch-level tokens can be generated, they are inevitably contaminated by global bias, which is detrimental to dense prediction. EVA-Clip [24] is an improved version of CLIP [10]. Due to various advanced pretraining enhancement techniques and a larger scale of data, these factors will bring improvements to the model performance, but the pretraining of VLMs is not the focus of our research.

3) Analysis on Fine-Tuning of Clip: To investigate the optimal fine-tuning solution for the RS scene, we analyze the effects of fine-tuning methods of the visual encoders of CLIP in Table IV. These solutions include *(Line 1):* without fine-tuning, *(Line 2):* fine-tuning the attention layers, *(Line 3):* fine-tuning query and key projections only, *(Line 4):* fine-tuning key and value projections only, and *(Line 5):* fine-tuning the full CLIP. All solutions of *(Lines 1–5)* is only for image encoders.

Some researches on natural scenes indicate that fine-tuning key and value projections usually performs best [11], [14]. However, we observed that fine-tuning the full CLIP yields the best performance in the RS scene. This may be caused by the scene gap between the pretraining and downstream data.

Furthermore, we attempted to fine-tune the text encoder *(Line 6)* or perform cross-attention *(Line 7)* to achieve image–text modality alignment. However, the results in Table IV reveal a critical flaw in these fine-tuning solutions: the text encoder is interfered with the noise from complex and changing semantic information in RS scenes. Consequently, the model's performance collapses to an unusable level. Our proposed Bias-CO successfully mitigates this issue, delivering excellent performance. These results provide quantitative evidence of the "model collapse" phenomenon and well demonstrate the effectiveness of our proposed Bias-CO.

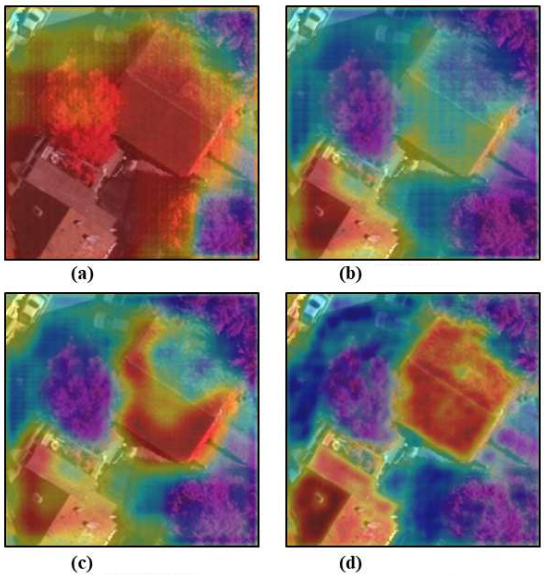


Fig. 7. *Visualization of the evolution of RS-Bias during training.* We show correlation weight maps that relate the text embeddings of "building" to image features refined by RS-Bias. The model is trained on the iSAID dataset for a total of 60 000 iters (iterations). (a) 0 iters. (b) 20 000 iters. (c) 40 000 iters. (d) 60 000 iters.

4) Analysis of Hyperparameters: We conduct the hyperparameter analysis in Table V to find the most effective combination of cutoff frequencies. It is observed that [9,7,3] achieves the optimal performance, which allows the low-frequency enhancement layer to effectively suppress harmful texture noise while achieving feature alignment within the high-dimensional semantic space.

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TABLE VI
HYPERPARAMETER ANALYSIS OF β ON iSAID. THE BEST-PERFORMING RESULTS ARE PRESENTED IN BOLD

β	VLM	DLRSD	Vaihingen	Potsdam	LoveDA	GaoFen-2
0.01	ConvNeXt-L	27.69	25.91	28.77	21.86	22.35
0.05	ConvNeXt-L	28.13	26.51	28.98	21.52	22.91
0.1	ConvNeXt-L	**28.52**	**26.76**	**29.26**	**21.97**	**23.85**
0.3	ConvNeXt-L	27.43	26.5	28.36	20.88	22.68
0.5	ConvNeXt-L	27.63	26.06	29.03	20.43	23.82
0.9	ConvNeXt-L	26.39	25.18	28.35	20.73	22.19

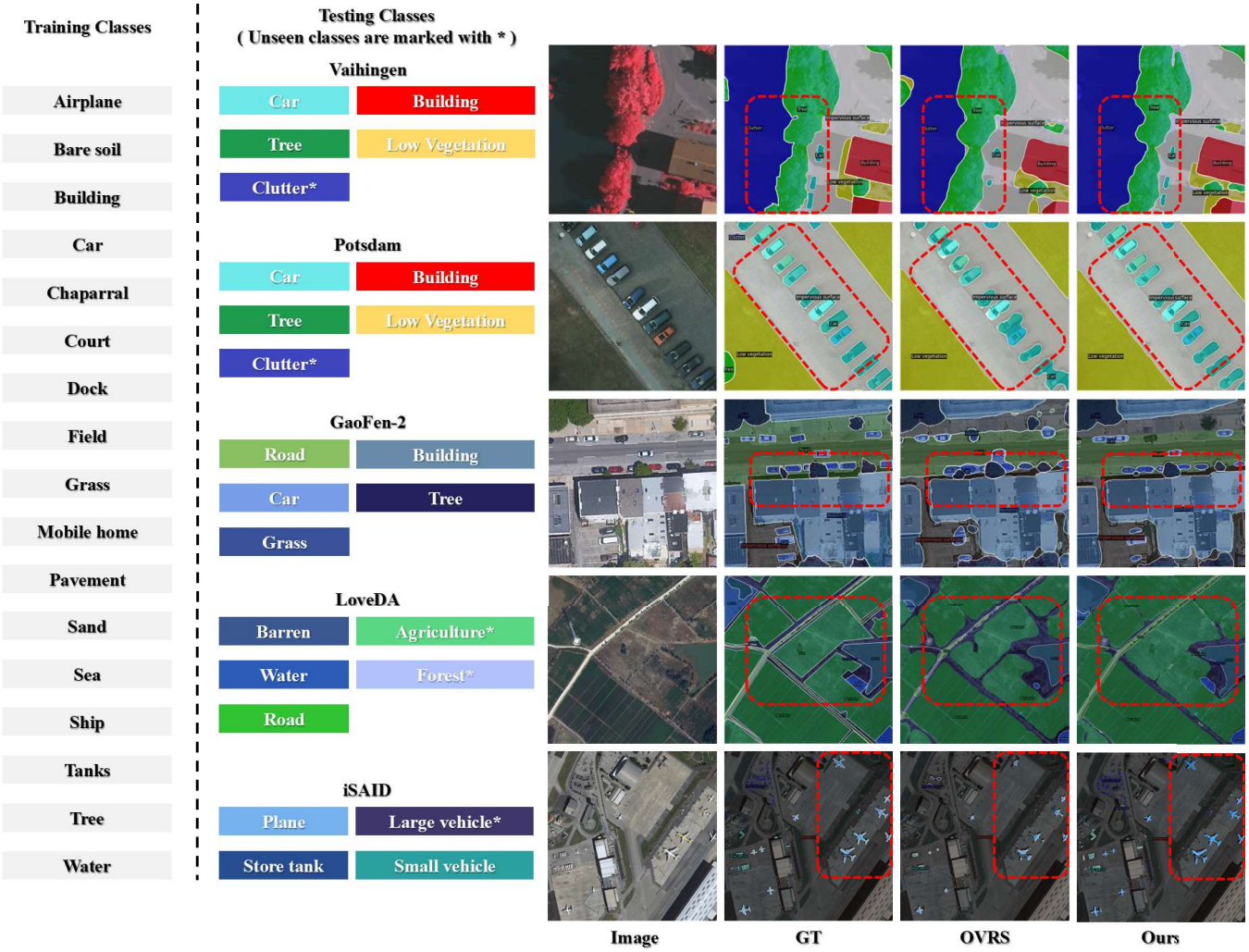


Fig. 8. *Qualitative comparison of the proposed method (trained in DLRSD).* (Left to Right) Input image, the ground-truth, and the results from state-of-the-art OVRS and our method. (Top to Bottom) Results of Vaihingen, Potsdam, GaoFen-2, LoveDa, and iSAID datasets. Some classes are named differently across various datasets (using synonyms). Despite the differences in nomenclature, the text embeddings are similar. We classify these categories as seen classes, such as "Bare soil" and "Barren."

The hyperparameter analysis of β is shown in Table VI. β is the update momentum that can influence the evolution of RS-Bias. We observe that using a value of 0.1 brings the best segmentation results. When β is too small, the RS-Bias struggles to effectively capture remote-sensing semantic representations, whereas excessively large β introduces noise and distorts the semantic information.

5) Evolution of RS-Bias: Fig. 7 demonstrates the analysis of how RS-Bias evolves during training. Fig. 7(a) shows the text–image correlation weights without any training. As similar textual descriptions can convey diverse and more complex semantic information in RS compared to natural scenes, the pretrained knowledge of the original CLIP is highly inaccurate. This results in a lack of correlation between image features and textual embeddings that share the same semantics.

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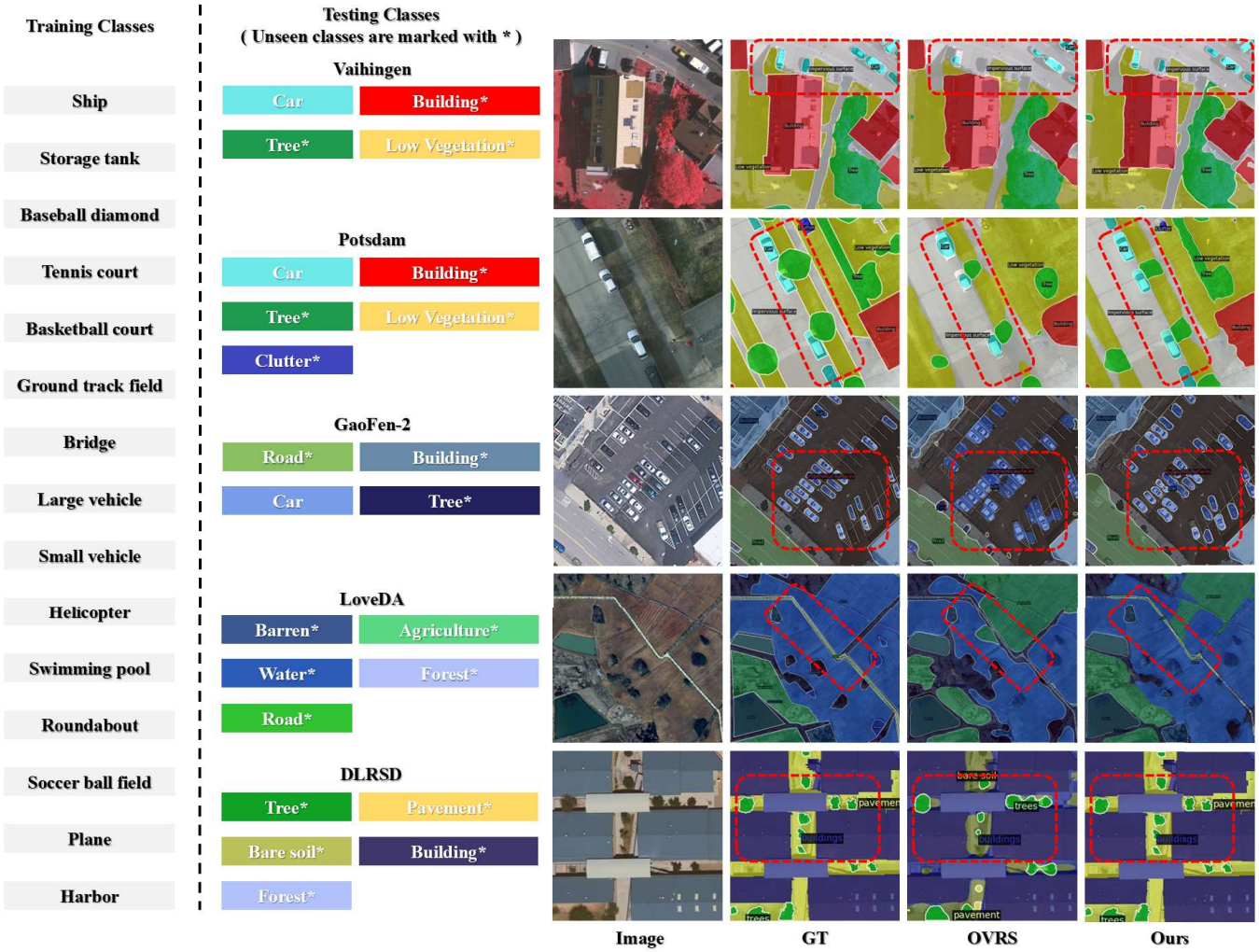


Fig. 9. *Qualitative comparison of the proposed method (trained in iSAID).* (Left to Right) Input image, the ground-truth, and the results from state-of-the-art OVRS and our method. (Top to Bottom) Results of Vaihingen, Potsdam, GaoFen-2, LoveDa, and DLRSD datasets. Some classes are named differently across various datasets (using synonyms). Despite the differences in nomenclature, the text embeddings are similar. We classify these categories as seen classes, such as "Small vehicle" and "Car."

As training progresses, RS-Bias gradually captures the key semantic information relevant to the RS scene and injects it into the modal. This positive evolution enables bidirectional interaction and tighter text–image alignment, which translates into increasingly accurate segmentation results in Fig. 7(b)–(d). In addition, our method can even attend to the building in the upper-left corner of the image, of which only a small portion is visible. This observation shows that RS-Bias can provide the model with reliable and accurate semantic information support when contextual cues are missing. These results consistently prove that RS-Bias can more accurately capture semantic representations during the training process.

E. Comparisons With State-of-the-Art Methods

We compare the performance of the proposed soft-guided open-vocabulary RS semantic segmentation framework with state-of-the-art semantic segmentation methods on six publicly available RS datasets: 1) iSAID [32]; 2) DLRSD [33], 3) LoveDa [34]; 4) Vaihingen [35]; 5) Potsdam [36]; and 6) GaoFen-2 [37]. These representative methods include: SCAN [53], SAN [6], MAFT [12], SED [14], CAT-Seg [11], SegEarth-OV [48], and OVRS [7]. The segmentation accuracies using mIoU are reported as shown in Table VII. The qualitative results of validation datasets are presented in Figs. 8 and 9. The results reported in Table VII are based on the full set of categories in the original datasets. For the sake of clarity, the visualization results only highlight the categories present in the displayed images along with their corresponding annotation colors.

As shown in Table VII, when trained on the DLRSD dataset, our model achieves a mIoU of 42.28%, 87.75%, 37.77%, 32.45%, 26.4%, and 30.06% on validation datasets, which surpasses any previous best method. Notably, compared to the state-of-the-art OVRS method, our approach shows substantial improvements of 3.44%, 2.52%, 4.03%, 3.46%, and 3.03% mIoU on iSAID, DLRSD, Vaihingen, Potsdam, and GaoFen-2. However, it falls short of the top performance on the LoveDA dataset compared to the OVRS method, though it still ranks second in mIoU. We also present qualitative results of validation datasets in Fig. 8, demonstrating the efficacy

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TABLE VII

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE TRAINING DATASET OF DLRSD OR iSAID. WE REPORT mIoU FOR EVERY METHOD. THE BEST-PERFORMING RESULTS ARE PRESENTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED. IMPROVEMENTS OVER THE SECOND-BEST ARE HIGHLIGHTED IN GREEN WHILE THE GAP TO THE BEST IS MARKED IN RED

Model	VLM	Training Dataset	iSAID	DLRSD	Vaihingen	Potsdam	LoveDA	GaoFen-2
SCAN$_{CVPR'2024}$ [53]	ViT-L/14	DLRSD	35.1	76.52	26.69	21.03	24.94	17.81
SAN$_{CVPR'2023}$ [6]	ViT-L/14	DLRSD	30.66	79.36	31.92	26.95	15.31	24.89
MAFT$_{NeurIPS'2023}$ [12]	ViT-L/14	DLRSD	19.28	83.47	20.87	<u>28.99</u>	23.97	26.61
SED$_{CVPR'2024}$ [14]	ConvNeXt-L	DLRSD	21.05	<u>85.23</u>	29.41	20.31	<u>26.81</u>	22.75
CAT-Seg$_{CVPR'2024}$ [11]	ViT-L/14	DLRSD	22.74	83.99	31.3	26.22	26.36	<u>27.03</u>
SegEarth-OV$_{CVPR'2025}$ [48]	ViT-L/14	-	23.48	17.9	19.77	18.95	19.5	12.43
OVRS$_{TGRS'2025}$ [7]	ViT-L/14	DLRSD	<u>39.04</u>	85.19	<u>33.74</u>	27.57	**27.19**	25.38
Ours	ConvNeXt-L	DLRSD	**42.28** (+3.44)	**87.75** (+2.52)	**37.77** (+4.03)	**32.45** (+3.46)	26.4 (-0.79)	**30.06** (+3.03)
SCAN$_{CVPR'2024}$ [53]	ViT-L/14	iSAID	70.61	17.09	9.88	19.41	13.95	11.55
SAN$_{CVPR'2023}$ [6]	ViT-L/14	iSAID	88.97	20.28	14.79	16.38	9.45	15.97
MAFT$_{NeurIPS'2023}$ [12]	ViT-L/14	iSAID	88.17	19.36	8.38	13.31	10.29	10.22
SED$_{CVPR'2024}$ [14]	ConvNeXt-L	iSAID	92.08	<u>25.45</u>	16.43	18.32	11.6	19.2
CAT-Seg$_{CVPR'2024}$ [11]	ViT-L/14	iSAID	92.35	22.13	14	15.79	<u>19.75</u>	12.71
SegEarth-OV$_{CVPR'2025}$ [48]	ViT-L/14	-	23.48	17.9	19.77	18.95	19.5	12.43
OVRS$_{TGRS'2025}$ [7]	ViT-L/14	iSAID	<u>93.41</u>	20.67	<u>20.82</u>	<u>19.94</u>	17.43	<u>19.39</u>
Ours	ConvNeXt-L	iSAID	**94.49** (+1.08)	**28.52** (+3.07)	**26.76** (+5.94)	**29.26** (+9.32)	**21.97** (+2.22)	**23.85** (+4.49)

and generalization capability of our proposed approach in comparison to the current state-of-the-art methods. In addition, when analyzing the segmentation results further, we observed some noteworthy outcomes. Although the metrics of our method are slightly lower than those of OVRS on the LoveDA dataset, this may be due to errors or inconsistent rules of labeling in the dataset. For example, in Fig. 8 Row 4, the labeling rule of "Road" is not clear; some small roads between "Agriculture" are labeled, while others are not. Moreover, our approach shows significant advantages when handling objects with extreme scales, such as the results of "Plane" in Fig. 8 Row 5. This capability is attributed to the comprehensive RS semantic knowledge provided by our proposed Bias-CO.

iSAID is a more challenging dataset because there is a potential dataset domain and category gap between iSAID and validation datasets, i.e., most of the classes during testing are not seen during training. This gap results in suboptimal performance of current methods. Our method demonstrates significantly superior segmentation performance for this dilemma. Specifically, our method with ConvNeXt-L significantly outperforms the previous best OVRS by 1.08%, 3.07%, 5.94%, 9.32%, 2.22%, 4.49%, reaching 94.49%, 28.52%, 26.76%, 29.26%, 21.97%, 23.85% mIoU on all five validation datasets. To the best of our knowledge, it is the current optimal performance on this benchmark. Fig. 9 shows the visualized comparison with state-of-the-art method OVRS. Although the testing classes are mostly unseen, our method can accurately segment both seen and unseen complex objects within diverse geographic landscapes, such as the small Car in Fig. 9 Row 3 and the narrow Road in Fig. 9 Row 4. Moreover, for original discriminative characteristics in natural scenes that may be absent or even opposite categories in the RS scene, such as "Tree", as shown in Fig. 9 Row 5, our model effectively corrects the feature discrimination modeling and accurately segments these objects.

F. Application Discussion

In this section, we discuss some advanced applications and potential optimization of our proposed soft-guided open-vocabulary RS semantic segmentation framework.

The first point is the applicability to other RS modalities. Adapting VLMs to downstream tasks presents an acknowledged bottleneck in the RS community. Our research addresses this critical issue. To validate the generalization of our proposed approach across other RS modalities, we conducted experiments on hyperspectral datasets SegMunich in Table VIII. Our method achieves a state-of-the-art performance of 34.14% mIoU, significantly surpassing existing methods. Furthermore, while our model framework is primarily designed for optical RS data, we suggest that users can also apply some optimization strategies based on modality characteristics to enhance the model's performance.

The second point is the scalability of the approach to larger vocabularies or more diverse RS applications. VLMs provide open vocabulary semantic segmentation models with a potential for generalization, expanding the vocabulary to a scale that closely resembles that of the real world. Our solution aims to ensure that this generalization ability experiences minimal loss when fine-tuning for specific scenes. Consequently, our proposed method is capable of handling scenes

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TABLE VIII
ADVANCED APPLICATIONS ON HYPERSPECTRAL MODALITY (SEGMUNICH). THE BEST-PERFORMING RESULTS ARE PRESENTED IN BOLD

Model	VLM	SegMunich
SED [14]	ConvNeXt-L [39]	30.06
CAT-Seg [11]	ViT-L/14 [10]	22.39
OVRS [7]	ViT-L/14 [10]	31.78
Ours	ConvNeXt-L [39]	**34.14**

with a larger vocabulary. Moreover, our proposed approach mainly focuses on the CLIP's encoders and image–text alignment. These contents are also critical considerations of other vision language adaptation tasks, such as RS object detection, video segmentation, or even medical image registration and industrial anomaly detection. Therefore, our proposed method has the potential to be applied to more diverse applications. Overall, we expect that the proposed method can shine in real applications. However, the deployment of such technology may lead to potential negative societal impacts, including the displacement of human labor by deep learning models, leading to job losses and increased unemployment rates.

The last point we should pay attention to is the inference speed limitation. Although our method does not significantly reduce the speed compared to the baseline methods, the inference speed of around 10 fps means that this method cannot be directly extended to fields such as video segmentation. This can be improved by pruning the model, thereby enhancing inference speed and reducing the number of parameters.

V. CONCLUSION

In conclusion, we proposed a soft-guided open-vocabulary RS semantic segmentation framework. Our framework comprises GC strategy, Bias-CO, and an improved upsampling decoder. We first utilize the GC strategy to implicitly guide the fine-tuning of CLIP's visual encoder. After that, we aggregate RS-Bias to infuse comprehensive RS knowledge into the model, thus implicitly aligning the image–text embedding space toward RS scenes. Finally, we employ an improved upsampling decoder to generate high-resolution cost maps for segmentation. Our method surpasses the previous state-of-the-art methods in standard benchmarks. This success underscores the promise and potential of our framework in advancing the field of open-vocabulary RS semantic segmentation.

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